



Python Programming - 2301CS404

Lab - 10

Roll No : 453

Name : Savaliya Dhruvi B.

01) WAP to read and display the contents of a text file. (also try to open the file in some other directory)

- in the form of a string
- line by line
- in the form of a list

```
In [5]: fp = open("file1.txt","r")
# print(fp.read())
# d = fp.readline()
# print(d,end = " ")
l = fp.readlines()
print(l,end = " ")
fp.close()
```

```
['Hello World!\n', 'This is Python Lab! ']
```

02) WAP to create file named "new.txt" only if it doesn't exist.

```
In [6]: fp = open("new.txt","w")
fp.write("This is Create New.txt.")
fp.close()
```

03) WAP to read first 5 lines from the text file.

```
In [14]: fp = open("file1.txt","r")
for i in range(0,5):
```

```
print(fp.readline(),end = " ")
fp.close()
```

Hello World!
 Welcome Python!!
 This is Python Lab!
 Python is High Level Language!
 Vishva Raiyani!

04) WAP to find the longest word(s) in a file.

```
In [3]: fp = open("file1.txt","r")
ml = 0
l_word = []

for line in fp:
    words = line.split()
    for word in words:
        l = len(word)

        if l > ml:
            ml = l;
            l_word = [word]
        # elif l == ml:
        #     l_word.append(word)
fp.close()
print("Longest word length:", ml)
print("Longest word(s):", l_word)
```

Longest word length: 9
 Longest word(s): ['Language!']

05) WAP to count the no. of lines, words and characters in a given text file.

```
In [ ]: fp = open("file1.txt","r")
count = 0;
l = fp.readlines()
print("line:",len(l),end = " ")
fp.seek(0)
print("word:",len(fp.read().split()))
fp.seek(0)
print("character:",len(fp.read()))

fp.close()
```

06) WAP to copy the content of a file to the another file.

```
In [27]: v = open("file1.txt","r")
r = v.read()
n = open("file2.txt","w")
n.write(r)
v.close()
n.close()
```

07) WAP to find the size of the text file.

```
In [4]: fp = open("file1.txt", "r")
content = fp.read()
fp.close()
size = len(content)
print("File size:", size, "bytes")
```

File size: 104 bytes

08) WAP to create an UDF named frequency to count occurrences of the specific word in a given text file.

```
In [10]: # import string

def frequency(filename, search_word):
    fp = open(filename, "r")
    count = 1

    for line in fp:
        # line = line.translate(str.maketrans("", "", string.punctuation))
        words = line.lower().split()
        for w in words:
            if w == search_word.lower():
                count += 1

    fp.close()
    return count

fname = "file1.txt"
search_word = input("Enter word to search: ")
result = frequency(fname, search_word)
print("Frequency of", search_word, "=", result)
```

Frequency of Hello = 2

09) WAP to get the score of five subjects from the user, store them in a file. Fetch those marks and find the highest score.

```
In [19]: # fp = open("file4.txt", "w")

# marks = []

# for i in range(1, 6):
#     score = int(input(f"Enter marks for subject {i}: "))
#     marks.append(score)
#     fp.write(str(score) + "\n")

# fp.close
fp = open("file4.txt", "r")

scores = []

for line in fp:
    scores.append(int(line.strip()))

fp.close()
```

```
highest = max(scores)

print("Highest score =", highest)
```

Highest score = 85

10) WAP to write first 100 prime numbers to a file named primenumbers.txt

(Note: each number should be in new line)

```
In [17]: fp = open("primeNumber.txt", "w")
start = 1;
li = []
n = 2;
while(len(li) < 100):
    for j in range(2,n):
        if(n % j == 0):
            break
    else:
        li.append(n)
        n+=1
print(len(li))
li=list(map(str, li))
l="\n".join(li)
fp.writelines(l)
fp.close()
```

100

11) WAP to merge two files and write it in a new file.

```
In [51]: f1 = open("file1.txt", "r")
f2 = open("file2.txt", "r")
f3 = open("file3.txt", "a")
f3.write(f1.read())
f3.write(f2.read())
f1.close()
f2.close()
f3.close()
```

12) WAP to replace word1 by word2 of a text file. Write the updated data to new file.

```
In [54]: f1 = open("file1.txt", "r")
f1.seek(0)
a = f1.read().replace("Python", "Python Programming")
f4 = open("file4.txt", "a")
f4.write(a)
f1.close()
f4.close()
```

13) Demonstrate tell() and seek() for all the cases(seek from beginning-end-current position) taking a suitable example of your choice.

```
In [60]: fp = open("file1.txt","r")
         fp.read(3)
         print(fp.tell())
         fp.seek(0,2)
         print(fp.seek(0,2))
         fp.seek(0,0)
         print(fp.seek(0,0))
         fp.close()
```

```
3
104
0
```

```
In [ ]:
```